



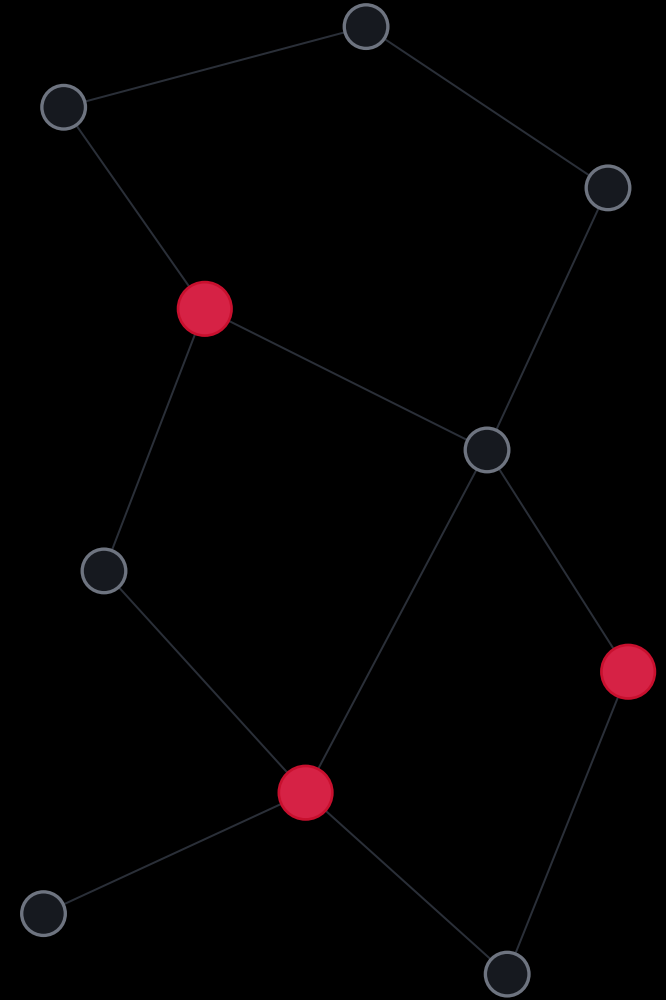
ARIC BROWN BAG · 29 SEPTEMBER 2026

From Planning Problems to Quantum Tests

How J1J turns operational requirements into models,
and tests the methods used to solve them.

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Who we are

JII at a glance

Mission

Making society computable to unlock humanity's potential

JII brings together experts in mathematical optimization, HPC, generative AI, and quantum technology to deliver optimization and QML services worldwide to a wide range of industries.



FOUNDED
2018



HEADQUARTERS
Tokyo, Japan



CEO
Yu Yamashiro



PRESENCE
Japan, UK,
EU, US

ABOUT J1J

Operational decisions, made computable

01

Model the decision

Translate industry knowledge, constraints and goals into mathematical models.

02

Build the workflow

Develop optimisation software and integrate it into planning and scheduling.

03

Test new methods

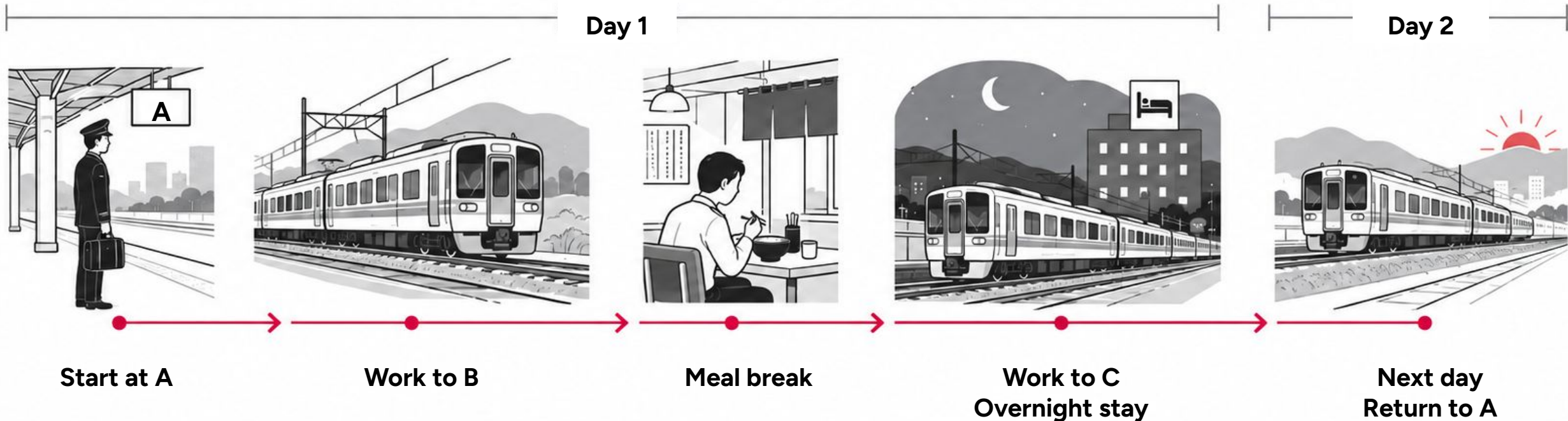
Develop and benchmark classical and quantum approaches to difficult problems.

TODAY: TWO COMPLEMENTARY EXAMPLES



Overview

What is a crew duty?



The challenge: combine many duties while respecting rules and practical needs.

THE STARTING POINT

A valid plan can still be a bad plan

Illustrative example: a duty meets the formal rules, but leaves an awkward meal break.

01

What can change?

Which trips belong in each duty?
Where can the crew change trains?

02

What must hold?

Working-time rules, available
connections and suitable rest
opportunities.

03

What makes it useful?

A plan must be workable for the
people who use it, as well as efficient.

MAKE THE REQUIREMENTS EXPLICIT

1 Choose a task → 2 Understand data → 3 Capture rules → 4 Set objectives → 5 Review plans

Turning railway expertise into a planning model

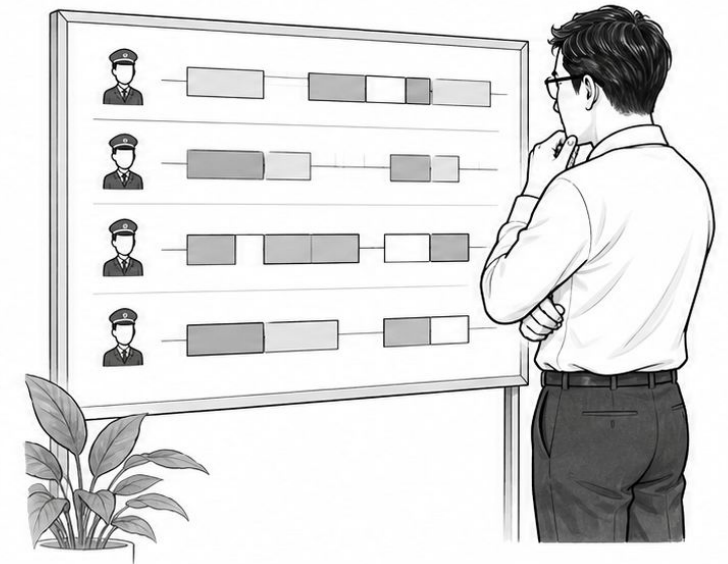
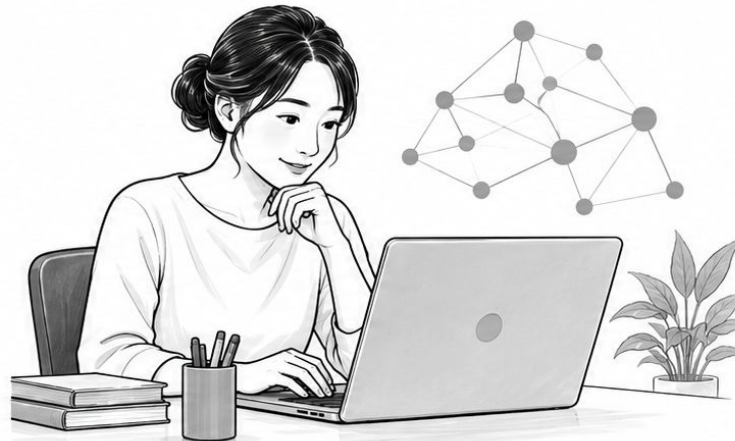
Railway planners



JIJ modelling



Proposed crew duties



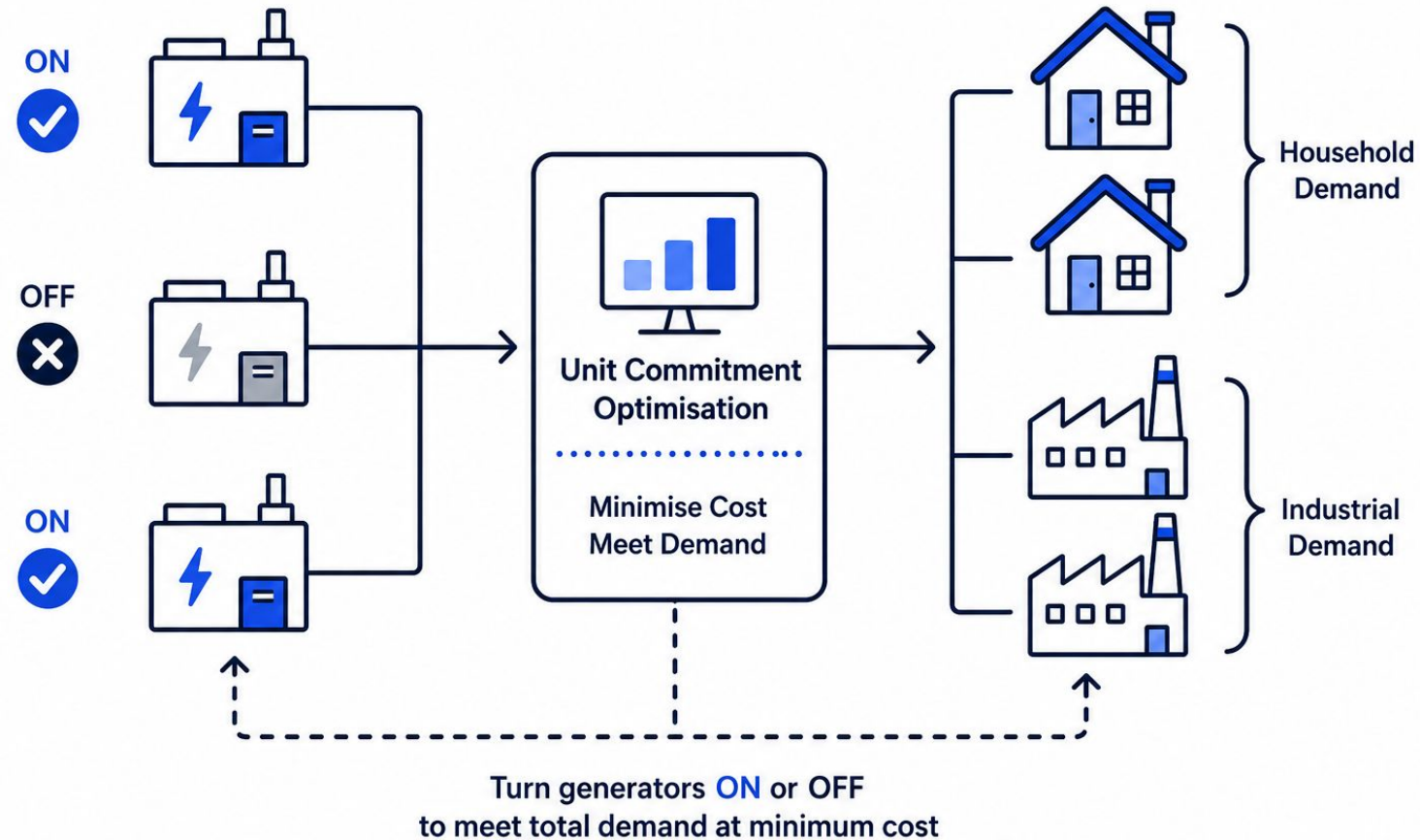
Planner feedback refines the model

Planners' feedback helps turn valid schedules into practical ones.

CASE 2 · SPARQ

Which generators should run, and when?

Unit Commitment Problem



CASE 2 · SPARQ

Which generators should run, and when?

Unit commitment combines on/off decisions with continuous power output.



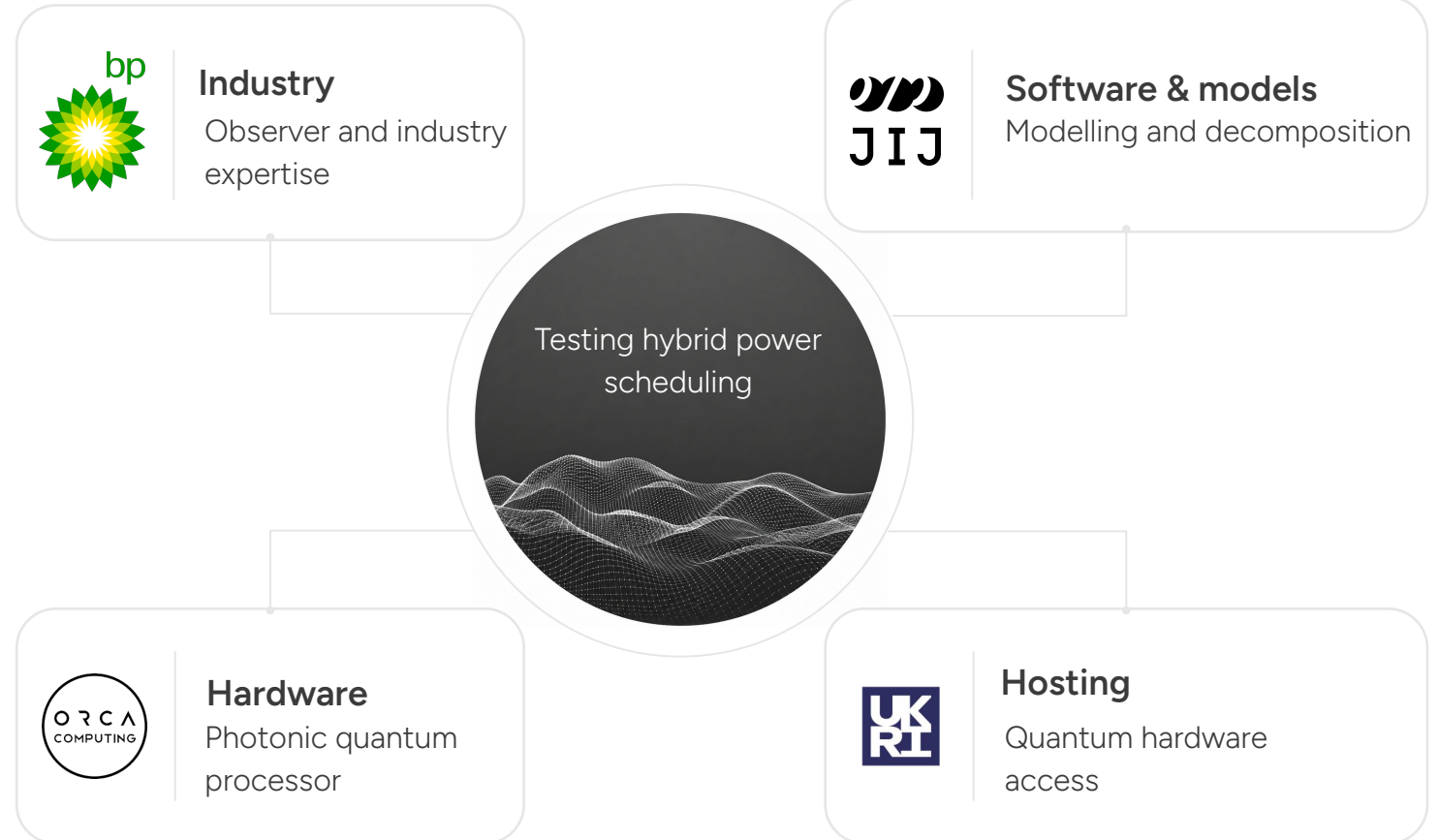
The decision

Which generators should run, when, and at what output? Meet demand at low cost.



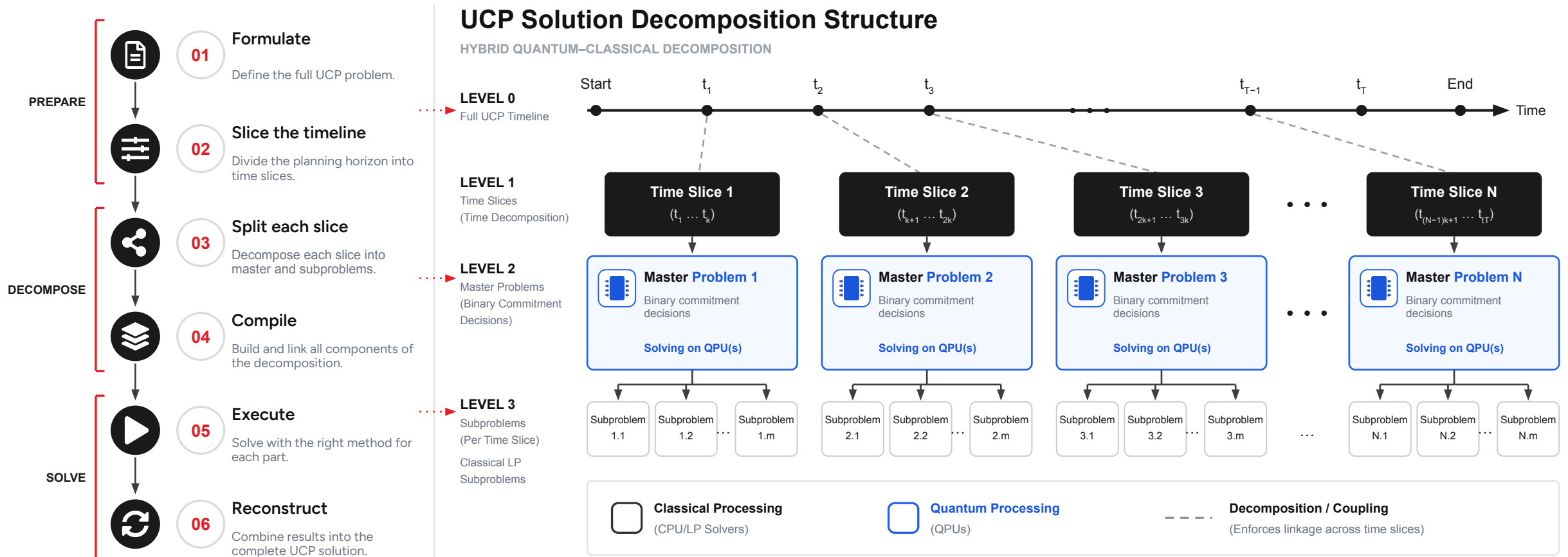
The constraints

Output limits, start-up conditions, reserves and operating requirements.



THE METHOD

Split the problem, send each part where it runs best



KEY CONCEPT:

Continuous dispatch
stays on classical LP solvers.

Binary commitment decisions
go to the quantum processor.

Each part runs on the technology
that solves it best.

HOW TO HANDLE THE PROBLEM SCALE

The processor takes 96 variables at a time

A direct mapping of an instance this size onto near-term photonic hardware is not slow. It is intractable.



32 qumodes

Resampling three times lifts each subproblem to 96 variables. Everything above that has to be decomposed.



2 delay loops

Limits how well the solver can train on each subproblem, which caps the quality it can reach.

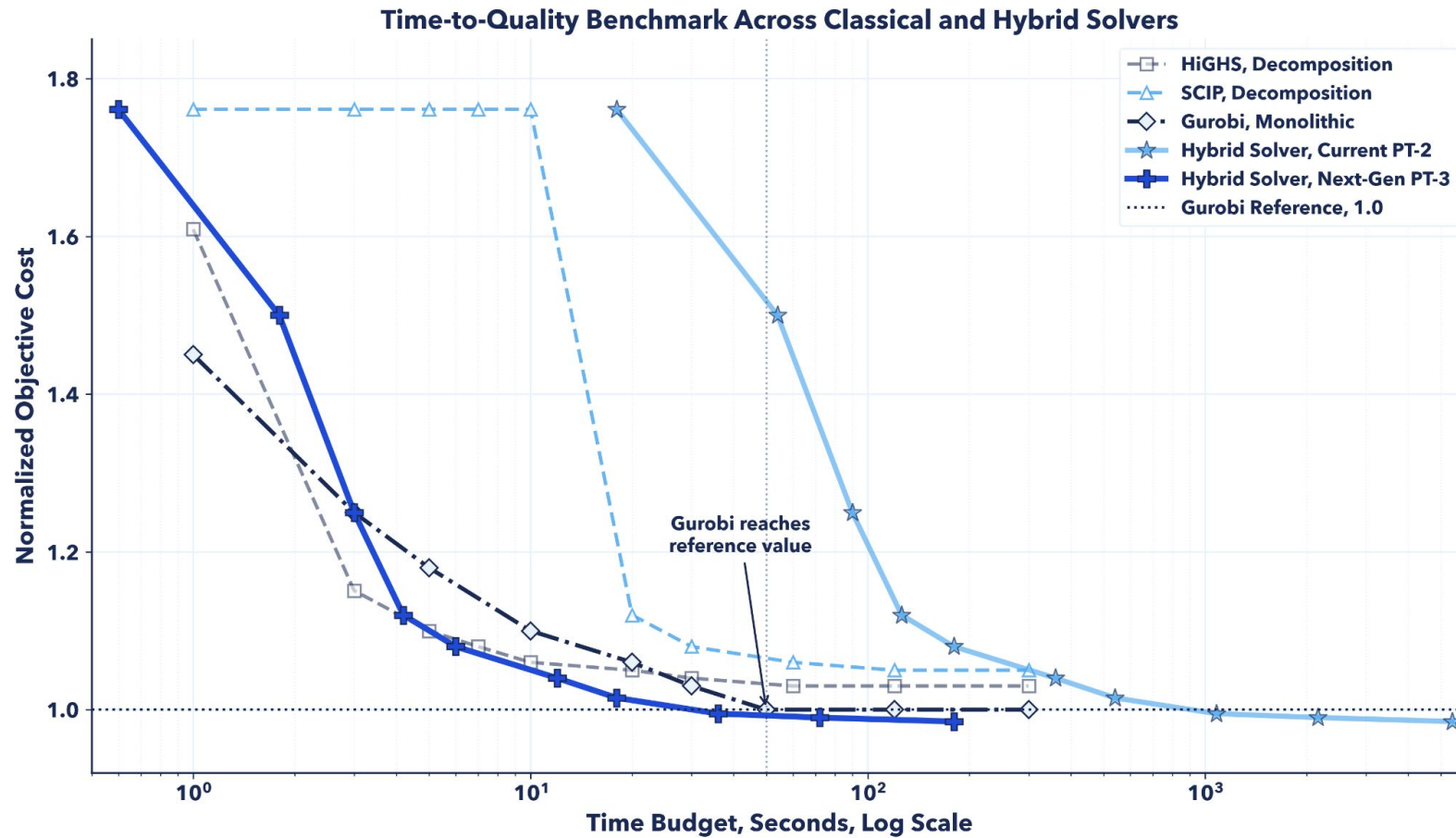


300 ms per batch

Sampling overhead dominates wall clock, and it is paid once per subproblem, every iteration.

Decomposition is not free. The smaller each subproblem, the less of the global structure the solver can see at once.

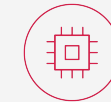
THE RESULT



Projection, not a measured result. Applies the published PT-3 specification to the measured PT-2 series.

**30x****faster sampling**

300 ms to 10 ms

**375****variables per execution**

up from 96

PT-3. Available by the end of 2026.

SPARQ · METHOD AND EVIDENCE

A hybrid workflow, tested against baselines

Largest benchmark: 25,755 variables and 48,939 constraints in the full problem.

01

Quantum component

J1J decomposes the model. Smaller on/off subproblems run on ORCA's processor.

02

Classical component

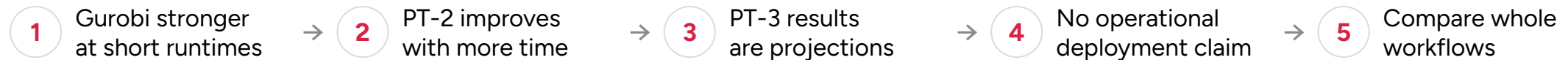
Classical solvers handle power output. The workflow reconstructs and checks the full plan.

03

What was demonstrated

Feasible full-problem solutions using real quantum hardware within a hybrid workflow.

BOUNDARY: BENCHMARK EVIDENCE, NOT COMMERCIAL QUANTUM ADVANTAGE



WHAT THE TWO CASES TEACH US

Different evidence, one disciplined approach

JR Central shows the modelling challenge. SparQ shows how to test a new solving approach.

01

Start with users

Make tacit knowledge and practical requirements explicit before judging a solution.

02

Match claims to evidence

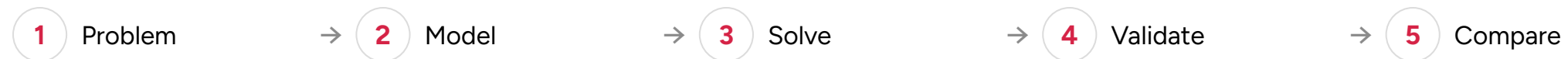
Customer testimony, benchmark results and deployed impact support different claims.

03

Compare fairly

Use the same problem, constraints and time budget. Separate measured results from projections.

THE PATH TO A CREDIBLE DECISION





Share your challenge

Where would you start?

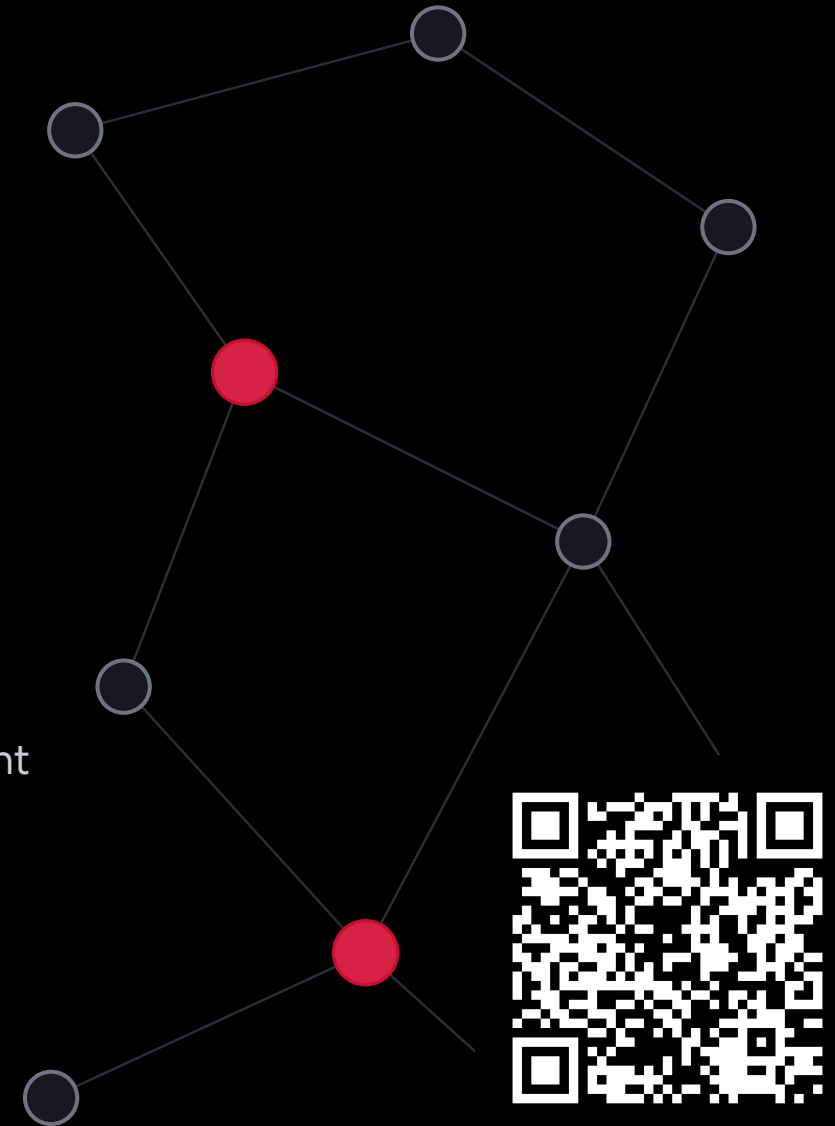
Which recurring planning decision would you improve?

What evidence would convince you that a new method is better?

Choose a task, agree success criteria and compare a pilot with your current approach.

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